

# CURRICULUM VITAE

## Shijie Xu

Faculty of Environmental Earth Science, Hokkaido University  
Kita-10 Nishi-5, Kita-ku, Sapporo, Hokkaido, 060-0810, Japan

✉ [shijie.xu@ees.hokudai.ac.jp](mailto:shijie.xu@ees.hokudai.ac.jp)  [0000-0001-6974-353X](https://orcid.org/0000-0001-6974-353X)



### Date of Birth

March 9th, 1995

### Education

|                                                                                                                             |            |
|-----------------------------------------------------------------------------------------------------------------------------|------------|
| Ph.D. in Environmental Science, Hokkaido University                                                                         | March 2025 |
| <i>Dissertation Title: "Accurate and Fast Predictions of Protein Structures and Properties via Protein Language Models"</i> |            |
| M.S. in Computer Science, University of Science and Technology of China                                                     | June 2020  |
| B.S. in Mathematics, University of Science and Technology of China                                                          | June 2017  |

### Research Experience

|                                              |                      |
|----------------------------------------------|----------------------|
| Postdoctoral Researcher, Hokkaido University | April 2025 - Present |
|----------------------------------------------|----------------------|

### Research Interests

Bioinformatics, Deep Learning, Computational Chemistry, Computer Science, Mathematics

### Skills

**Programming:** Python / C++ / CUDA /  $\text{\LaTeX}$  / FORTRAN / Rust / Mathematica / MATLAB / Coq / Haskell

**Web development:** HTML / JavaScript / FastAPI / Flask / Django / Go

**Software:** AutoDock Vina / PyMOL / Git / Docker / Singularity / VSCode

**Hardware:** GPU workstation building / Server maintenance / Raspberry Pi

**Languages:** English (fluent), Japanese (basic), Chinese (native)

### Publications

1. [Shijie Xu](#), Ke Xu, and Akira Onoda\*. "Accurate Prediction of Enzymatic Degradation of Plastics by Language Models." *Chemistry Letters* 55(2): upag022, 2026, DOI: <https://doi.org/10.1093/chemle/upag022>

2. Shijie Xu and Akira Onoda\*. “Probe-Based Identification of Metal-Binding Sites Using Deep Learning Representations.” *bioRxiv* (Preprint), **2025**, DOI: <https://doi.org/10.1101/2025.10.04.680417>
3. Shijie Xu and Akira Onoda\*. “Accurate and Rapid Prediction of Protein pK<sub>a</sub>: Protein Language Models Reveal the Sequence-pK<sub>a</sub> Relationship.” *Journal of Chemical Theory and Computation* 21(7): 3752–3764, **2025**, DOI: <https://doi.org/10.1021/acs.jctc.4c01288>
4. Shijie Xu and Akira Onoda\*. “PsiPartition: Improved Site Partitioning for Genomic Data by Parameterized Sorting Indices and Bayesian Optimization.” *Journal of Molecular Evolution* 92: 874–890, **2024**, DOI: <https://doi.org/10.1007/s00239-024-10215-7>
5. Shijie Xu and Akira Onoda\*. “Accurate and Fast Prediction of Intrinsically Disordered Protein by Multiple Protein Language Models and Ensemble Learning.” *Journal of Chemical Information and Modeling* 64(7): 2901–2911, **2024**, DOI: <https://doi.org/10.1021/acs.jcim.3c01202>

## Presentations

- Shijie Xu and Akira Onoda, “Accurate and Fast Prediction of Metal-binding Sites in Proteins Using Deep Learning Methods”, 52nd Biomolecule Science Symposium, 14th May - 16th May 2026, Himeji, Japan. [Oral]
- Shijie Xu and Akira Onoda, “Accurate and Fast Prediction of Protein pK<sub>a</sub> by A Protein Language Model”, the 19th Symposium on Biorelevant Chemistry, 2nd Sep - 4th Sep 2025, Kyoto, Japan. [Poster]
- Shijie Xu and Akira Onoda, “Accurate and Fast Prediction of Protein pK<sub>a</sub> by Protein Language Model”, 51st Biomolecule Science Symposium, 26th Jun - 27th Jun 2025, Sapporo, Japan. [Oral]
- Shijie Xu and Akira Onoda, “Accurate and Fast Prediction of Protein pK<sub>a</sub> by Protein Language Model”, 8th ICRDD International Symposium, 22nd Oct - 24th Oct 2024, Sapporo, Japan. [Poster]
- Shijie Xu and Akira Onoda, “Accurate Prediction of Intrinsically Disordered Protein by Protein Language Models”, the 24th Annual Meeting of the Protein Science Society of Japan, 11th Jul - 13th Jul 2024, Sapporo, Japan. [Poster]
- Shijie Xu and Akira Onoda, “Accurate Prediction of Protein pK<sub>a</sub> by Geometric Deep Learning”, the 104th CSJ annual meeting, 18th Mar - 21st Mar 2024, Funabashi, Japan. [Oral]
- Shijie Xu and Akira Onoda, “Accurate Prediction of Intrinsically Disordered Protein by Protein Language Models”, the 17th Biorelevant Chemistry Symposium, 8th Sep - 10th Sep 2023, Chiba, Japan. [Poster]
- Shijie Xu and Akira Onoda, “Fast and Accurate Prediction of Intrinsically Disordered Protein by Protein Language Model”, the 103rd CSJ annual meeting, 22nd Mar - 25th Mar 2023, Chiba, Japan. [Oral]

## Awards, Fellowships, and Grants

- Program for Supporting Challenging and Interdisciplinary Research Field, 2022, 2023, 2024
- Hokkaido University DX Doctoral Fellowship, 2022-2025
- Recipient of the Outstanding Student Scholarship, University of Science and Technology of China, 2016

- S.-T. Yau College Student Mathematics Contest, Beijing, China, 2016
- Recipient of the Outstanding Freshman Scholarship, University of Science and Technology of China, 2014
- Winner of the First Prize in the National High School Mathematics League (province level), Hefei, Anhui, China, 2012